

5. Project Code (DAX Queries)

DAX Queries Explanation:

DAX (Data Analysis Expressions) is a formula language used in Power BI to define custom calculations for your reports and dashboards. In this section, we'll provide some important DAX queries that are used for key metrics and explanations.

Change Date to Correct Format:

One important DAX formula is for **date transformation**. If the source data has inconsistencies in date formats, a DAX formula can standardize them:

dax

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Corrected Date =

DATEVALUE([Raw Date Column])

This formula ensures that any raw date input is converted into a standard date format recognized by Power BI.

Total Revenue:

To calculate the **Total Revenue** from sales data, the following DAX query can be used:

dax

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Total Revenue = SUM('Sales'[Revenue])

This simple DAX query sums the revenue column in the Sales table to calculate the total revenue.

Monthly Profit:

To calculate monthly profit, you would need to subtract costs from revenue. Here's a sample DAX expression:

dax

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Monthly Profit =

CALCULATE(
SUM('Sales'[Revenue]) - SUM('Sales'[Cost]),
MONTH('Sales'[Date]) = MONTH(TODAY()))

)

This query calculates the monthly profit by summing revenue and subtracting costs for the current month.

Yearly Revenue by Region:

For revenue segmented by region over the years, you might write something like this:

dax

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Yearly Revenue by Region =

```
CALCULATE(  
    SUM('Sales'[Revenue]),  
    ALLEXCEPT('Sales', 'Sales'[Region], 'Sales'[Year])  
)
```

This sums up the revenue by region and year, excluding other filters.